

PHILOSOPHY / EDUCATION / AI RESEARCH

The Art of Knowing What Matters

Wisdom, Education
and Artificial Intelligence

FROM TRANSMITTING INFORMATION
TO CULTIVATING JUDGMENT

Featuring the PHRONESIS Research Program
Distilling and Validating Contextual Judgment

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A Note to the Reader

This book begins with an asymmetry: we can place the same information before a million people without a million people acquiring the same capacity to use it. And a new technical capacity does not decide by itself whether it is worth using, for what purpose, and with what consequences for others. Between the circulation of information and the formation of judgment lies a distance that communication speed does not erase.

We will call practical wisdom a contextual capacity to judge what is worth pursuing, to use critically what we know, and to revise our conduct in light of consequences, uncertainty, and the legitimate interests of those affected. This is a debatable working definition, not the discovery of an eternal essence or a test of a person's worth. Contemplative, religious, or existential wisdom exceeds this definition; the book encounters it without claiming to exhaust it.

The first two parts follow the human problem: what can be transmitted, and how education and institutions might foster judgment. The last two investigate the technical problem: whether we can distill structures from texts that help people make better decisions without confusing elegance of expression, majority agreement, or compression with wisdom.

Numbered references lead to the bibliography at the end. The documentation represents a critical exploration of primary sources and research syntheses, verified through September 10, 2026, not an exhaustive systematic review. The educational proposals and PHRONESIS program are the book's reasoned constructions. They are not validated interventions and do not describe empirical results already obtained. The illustrative calculation in appendix B verifies only a constructed micromodel. Scenarios without bibliographic references are invented examples used to test ideas.

Prologue. The School That Knew Too Much

Imagine a school introducing a digital participation system. Students answer questions, receive points, and teachers immediately see who has contributed and who has fallen behind. The program works. The monthly report shows more answers, faster homework submissions, and fewer moments when the class seems inactive. The principal receives congratulations.

At a meeting, a teacher asks whether students understand better. There are data on their answers, the computer specialist says. There is even a model predicting who will receive a low grade. The teacher rephrases: if we remove the platform tomorrow, can students solve a new problem? That question was not in the report.

A student says he answers quickly to avoid losing points, even when he needs time to think. A girl speaks less because she formulates answers at a different pace. Another student has discovered how to appear active without reading. None disputes that the platform collects accurate information. They dispute what it has been decided that information should mean.

It would be too simple to end the story by condemning technology. The platform can identify a real difficulty and save teachers time. It would be just as simple to condemn teachers for not defining enough metrics. Every new measure selects other details and leaves others outside the frame. The problem is not choosing between measurement and intuition, but the relationship between what we measure and what we try to cultivate.

In this imaginary school, information answers “what was recorded?” Usable knowledge lets us check an indicator, change an exercise, and design a test. Wisdom begins when we ask whether the objective pursued represents what matters, who bears the costs, and what should lead us to change direction.

The principal need not become a professional philosopher. They must be able to say: “We confused visible activity with learning. Let us keep what helps, stop what harms, and check separately.” Such a sentence is easy to write. It is harder for it to exist in an institution where its author is not punished for admitting an error.

This will be the book’s question: what should we build around a good sentence for it to become a real capacity of people and organizations?

1. Three Ways of Knowing

From a Message to a Capacity

Information can be copied without the sender losing it. A statement, table, file, or instruction can circulate almost identically among recipients. This is a property of representation, not a guarantee of understanding. For a reader unfamiliar with the language, a treatise on mechanics and an arbitrary string of signs may be equally useless in practice.

Here, information will mean communicable content that can change what a recipient believes about a situation. Usable knowledge requires more: sufficiently accurate representations, procedures, and skills enabling appropriate explanations, predictions, or actions. Having seen how a theorem is proved is insufficient; you must recognize the conditions under which the proof applies and reconstruct the relevant steps.

The difference can be observed in a programming example. “Passwords are not stored in plaintext” is information. Designing and verifying an authentication mechanism requires usable knowledge. Deciding what data should not be collected, what autonomy users should retain, and when a commercial feature creates disproportionate risks requires another level of judgment. The example is not a security specification; it shows that technical success does not by itself establish a purpose’s legitimacy.

Knowledge is not merely a collection of instructions. It includes the ability to distinguish cases, detect an error, and adapt a procedure. Yet even remarkable knowledge can be narrow. A specialist can perfectly optimize a poorly chosen objective. They can solve the assigned problem without noticing that its formulation excludes precisely the person who will bear the solution’s cost.

Wisdom, in the book’s working sense, also concerns relationships between purposes. It asks not only how to obtain more, but more of what, for whom, and in exchange for what. It can lead to acceleration,

slowing down, or refusal. It is identified with neither efficiency nor caution. Refusing to intervene in an injustice can lack discernment just as much as a rash intervention.

These distinctions do not create three sealed compartments. A statement can contain an instruction and a reflection on its purpose. A practical skill can include implicit moral judgment. A prudent decision can become impossible for lack of elementary information. We have dimensions supporting one another rather than a ladder on which each level replaces the preceding one.

Wisdom Is Not a Surplus of Intelligence

Instrumental intelligence can identify ingenious means to a given outcome. Practical wisdom, as proposed here, includes willingness to question the outcome itself. This distinction is conceptual; it does not imply that intelligence and wisdom are psychologically independent. Nor does it justify the convenient idea that lack of competence is a virtue.

Suppose a team must reduce a service's response time. One solution removes difficult cases from the statistics. Another automates simple requests and reserves the saved time for difficult cases. The first may improve the indicator more. The second may better serve the institution's purpose. Discernment consists in noticing that the indicator is not the service's beneficiary.

Nor is professed kindness sufficient. Someone can sincerely wish others well and cause predictable harm because they do not understand the situation's mechanisms. Conversely, a very clear-sighted person can use that lucidity to exploit. The proposed definition links two requirements: adequacy to reality and justifiable responsibility toward those affected. The connection is normative and must be defended, not hidden beneath a technical score.

An objection immediately arises: who decides which interests are legitimate? No simple procedure eliminates every conflict. But the absence of an infallible arbiter does not make all positions equivalent. We can demand evidence for claims, reciprocity in justifications, ac-

cess to appeal, and attention to consequences. We can distinguish real disagreement from lying and authentic uncertainty from using uncertainty as a pretext.

Wisdom is not the reputation for being wise either. Calm discourse can protect an error. An awkward warning, inelegantly expressed, can identify the decisive danger. Assessment must move from the impression a person produces to the relationship between argument, context, and action.

We will therefore use three different questions. What does the message contain? What can the recipient do with it? What kind of judgment do they display when deciding whether and how to use it? Confusing these questions begins many educational illusions and many exaggerated promises about AI.

2. Philosophy as an Exercise in Judgment

When the Rule Cannot Replace the Person

In Phaedrus, Plato stages a problem surprisingly close to digital education: a text can circulate, but it does not know whom it addresses and cannot answer objections by itself. The criticism does not cancel writing's value; the dialogue itself reached us through writing. It separates memory's external medium from the activity through which someone comes to understand and answer for what they assert. [1]

For our problem, the lesson is structural. An identical message reaches people with different experiences, fears, and purposes. One needs a definition, another a counterexample, another reasons to abandon an interpretation benefiting them. Making content available is not equivalent to organizing the encounter between that content and the mind that will use it.

In Book VI of the Nicomachean Ethics, Aristotle distinguishes forms of knowledge and intellectual virtue. *Phronēsis* concerns practical deliberation about the good life and requires attention to a situation's particulars; *sophia* has another object, connected to theoretical wisdom. This distinction prevents us from treating every expertise or contemplation as sufficient capacity to decide well in concrete life. [2]

We need not adopt Aristotle's entire social universe to use the distinction. We can accept that rules do not apply themselves while rejecting a rigid distribution of authority among people. A contemporary philosophy of judgment must allow affected people to challenge how someone else claims to know their good.

A fruitful tension arises here. Without rules, every situation risks becoming arbitrary improvisation. With absolutized rules, relevant differences disappear. Discernment does not mean sitting comfortably in the middle, but knowing why we retain a rule and what kind of fact would justify an exception. A good exception must be explainable to someone other than its beneficiary.

Take a promise of confidentiality within a team. “Always keep the secret” can protect trust; it can also conceal serious danger. “Always tell everything” can defend transparency; it can also expose someone unnecessarily. Philosophy becomes useful when it asks us to identify the conflict, not when it offers a maxim seeming to eliminate it.

Reflection Before, During, and After Action

Another resource is the Buddhist text known as the Instructions to Rāhula, MN 61. It proposes examining actions before starting, during their course, and after completion, with attention to effects on oneself and others. This is a precise example from one tradition, not a description of all Asian philosophies. [3]

We can reconstruct this structure in secular terms: anticipation, monitoring, revision. Before acting, we formulate what we believe will happen. During action, we watch for signals incompatible with expectations. Afterward, we examine the outcome without retrospectively rewriting the story to make ourselves seem clairvoyant. This reconstruction is our pedagogical proposal, not an identity between a religious text and experimental method.

In *How We Think*, John Dewey treats reflection as investigation of a belief’s grounds and implications, not merely ideas passing through the mind. The relationship between doubt, inquiry, and experience provides a foothold for educating judgment: students must learn how a difficulty becomes an investigable question. [4]

There are important differences between these sources. They do not speak of the same good life, metaphysics, or social organization. The point of comparison is not to claim all traditions say the same thing. It is to extract a shared problem: how do we connect understanding to a practice open to correction?

The book proposes considering philosophy also as a discipline of moving between perspectives. The agent’s perspective shows intention and difficulty. The affected person’s perspective shows cost. The observer’s perspective may reveal contradiction. The future’s perspective asks which options we leave open. None automatically holds the verdict; together they can expose what a single position hides.

This does not demand endless deliberation over every gesture. That would caricature wisdom. It demands habits that make attentive thought available when the situation calls for it. Just as musicians practice slowly to perform fluently, judgment can be deliberately trained to become less dependent on impulse or memorized formulas.

Philosophy understood this way is not a shelf of prestigious answers. It is a workshop where reasons become visible, purposes debatable, and authority revisable. Precisely for this reason, it can play a central role in a world with almost instantaneous access to answers.

3. Why Experience Cannot Be Copied

The Context a Statement Loses

A mentor says: “Do not confuse quick success with the right direction.” The apprentice can repeat the sentence perfectly. But when an opportunity offers prestige, money, or belonging, that same sentence may become invisible. Not because it disappeared from memory, but because it is not activated in the appropriate situation or conflicts with a stronger motivation.

The difficulty of transmitting wisdom has at least three distinct levels. A representation of the problem must be transmitted. The capacity to recognize it in new situations must be developed. Willingness to act according to judgment must be sustained, even at a cost. A book works directly on the first level, indirectly on the second, and only very limitedly on the third.

There is another difficulty: someone’s conclusion hides the comparisons through which they reached it. An experienced person may see relevant evasion in an interlocutor’s sentence because they have encountered many variations of that situation. Transmitting only the verdict eliminates precisely the differences enabling it. The recipient receives the result of compression, not the criterion by which details were selected.

To transmit more faithfully, mentors should recount not only what they did, but which alternatives they had, what they did not know, what clue changed their mind, and when the same advice would have been wrong. Not every experience can be fully recounted. But the account can preserve judgment’s branching points instead of turning life into a sequence of inevitable conclusions.

This explains literature’s possible pedagogical value. A story can hold incompatible motives together without resolving them immediately. Readers can follow how a decision reasonable from one perspective

becomes burdensome from another. This is a conceptual justification for using narrative; it does not prove every novel improves discernment or fiction automatically produces empathy.

Experience, however, is not an infallible authority. It can repeat the same misinterpreted situation, exclude cases contradicting the lesson, and turn luck into personal merit. Saying “I lived through this” contributes a source of information; it does not close discussion. Experience gains value for others when it can be examined, compared, and corrected.

To Understand, to Be Able, and to Dare

A student can understand the argument for honesty yet anticipate an unjust penalty for admitting an error. An employee can identify a risk while depending economically on the person ignoring it. Describing these situations as an individual lack of wisdom shifts attention from constraint to character. Sometimes the problem is not that people do not know, but that the institution makes good conduct excessively costly.

Pedagogy must therefore not promise that every conflict can be resolved through personal development. It must develop discernment and make intelligible the structures obstructing it. Sometimes reasonable action consists in building support, documenting a fact, or seeking a safe avenue of challenge rather than immediate heroism. In other situations, delay perpetuates harm. The conditions make the difference.

It does not follow that we should cause suffering to produce wise people. That would be both a logical error and a dangerous pedagogy. Someone’s learning after a loss does not imply that the loss was necessary, sufficient, or justifiable. Schools can offer simulated conflicts, proportional responsibilities, and reparable failures without manufacturing trauma.

Nor should the emotional dimension be treated as a residue eliminated to obtain pure reason. In the model proposed here, emotion may signal a relationship’s or loss’s importance, but the signal must be in-

terpreted. A situation's outraging me does not by itself establish my description's truth; its not affecting me personally does not prove it lacks gravity.

Transmitting wisdom thus requires alternating proximity and distance. Proximity prevents turning people into indifferent variables. Distance makes examining our own interest and reaction possible. Education must create opportunities for both without turning personal disclosure into a public obligation.

We can now formulate why information circulates more easily than wisdom. A message can be externalized almost entirely; competence must be reconstructed in the recipient; practical judgment must be reconstructed in a relationship between recipient, situation, motivations, and institutions. This is a functional explanation, not a universal measurement of difficulty. Some information is extremely hard to understand, while some simple warnings can change a life immediately.

What remains stable is that successful transmission cannot be read from the message alone. It is read in the capacity the other person acquires and how it withstands a situation the teacher could not fully anticipate.

4. What the Science of Wisdom Can Measure

Multiple Instruments, Not a Universal Thermometer

The psychology of wisdom has no single accepted instrument measuring a unified property without controversy. Judith Glück and colleagues' comparative study examines differences between several instruments, including self-assessments and evaluations of responses to life problems. Some emphasize reflection and compassion; others experience, emotional regulation, or pragmatic knowledge of life. The choice of instrument influences what we can conclude. [5]

This does not mean the field is useless. It means we must ask what each method observes. A questionnaire can show how people describe themselves. A written response can show which considerations they can formulate in a test situation. Repeated observation can show how they behave in a particular environment. None of these observations alone authorizes a verdict on their entire character.

We will distinguish three objects of assessment. The first is a response's content: does it include relevant perspectives, uncertainties, and conditions? The second is task performance: does the person detect information changing the decision and adjust their conduct? The third is a disposition's stability: can they do this across contexts, especially when their own interest is involved? Moving from the first to the third is a hypothesis requiring verification.

The distinction also matters when assessing AI. A model can generate an excellent reflection on self-knowledge without maintaining the same conduct in a conversation under social pressure. A person can write an unremarkable answer and possess excellent practical judgment. Assessing language must not become a cultural tax on those who do not use the evaluator's preferred register.

There is also the problem of implicit values. An instrument always rewarding compromise will consider justified opposition to abuse less wise. An instrument always rewarding certainty will penalize ac-

knowledging limits. But an instrument rewarding every hesitation will confuse indecision with epistemic humility. Items must include situations in which the same verbal form becomes appropriate or inappropriate.

Instead of a universal ranking of people, we propose local performance profiles accompanied by explanations and limits. For example: the person identifies costs to others well but insufficiently distinguishes hypothesis from fact; or recognizes uncertainty but does not specify what information would reduce it. These findings are more useful for learning than the label “wise” or “unwise”.

The Situation May Matter More Than Reputation

Grossmann and Kross studied the asymmetry between judging our own problems and others’ problems. In their experiments, distancing perspective reduced this asymmetry in reasoning about close relationships. The result provides a basis for exercises reflecting from an observer’s perspective; it does not demonstrate that a brief technique produces a stable virtue in every domain. [6]

From this finding, we can construct a modest educational hypothesis. Before defending a controversial decision, students can reformulate it as if advising another person. Then they return to their own position and note what changed. The aim is not to cancel self-interest but to observe how it organizes attention.

Culture introduces another difficulty. A respectful, indirect response can be misinterpreted as lacking a position; a direct response can be interpreted as aggression. Comparing groups would require checking whether the instrument measures the same construct and whether scenarios are intelligible in different contexts. We will not infer societies’ wisdom from stereotypes about their antiquity, spirituality, or individualism.

We can nevertheless investigate differences without total relativism. If two communities interpret a family obligation differently, the data should preserve reasons, not just choice frequencies. We can separately analyze agreement on facts, differences between norms, and the

distribution of costs. Normative disagreement must not automatically be treated as measurement error; a false factual claim does not become true through cultural acceptance.

Decision quality must also be separated from its outcome. A well-justified decision can have a bad outcome because of an unpredictable event. An imprudent decision can get lucky. The proposed assessment therefore requires prospectively recording reasons and expectations, then examining the outcome. This does not eliminate subjective judgment, but limits the advantage of a story constructed after the fact.

Wisdom becomes investigable not when we find a number enclosing it, but when we formulate precise observations capable of contradicting a claim. Does a system claim to consider context? Let us change a decisive condition. Does it claim to recognize limits? Let us give it a situation where information does not permit a verdict. Does it claim to be fair? Let us examine who disappeared from its analysis.

5. The Pedagogy of Transfer

Knowledge Must Become Available

An educational reform oriented toward wisdom can begin with an error: disdaining memorization, teacher explanation, and repeated practice. Without accessible concepts and elementary skills, students have nothing with which to compare a new situation. Information's being findable does not mean a person knows what to search for, when to search, or how to recognize a wrong answer.

Dunlosky and colleagues' synthesis assigns high utility to retrieval practice and practice distributed over time. Self-explanation and interleaving problem types have more condition-dependent support. Summarization is not declared useless, but its benefits generalize less well across all tasks and learners. These findings concern learning, not proof of moral transformation. [7]

Our proposal is to preserve these foundations and organize them for contextual availability. Students must not merely reproduce the definition of a statistical mean, but recognize when the mean hides a relevant distribution. They must not merely know what a source is, but notice when ten texts repeat the same unverified claim.

One possible teaching sequence begins with explicit explanation and a worked example. A similar example follows, solved with support. Then comes a contrasting case where a changed condition makes the method inappropriate. Finally, students receive a new situation without the label of the chapter it comes from. Each step serves a different function: understanding, practice, discrimination, transfer.

Take the rule "a larger sample provides a more stable estimate". The teacher explains the idea and conditions. The class then compares two imaginary surveys: one small but well selected, the other large and composed solely of volunteers from an interested community. The useful lesson is not that size does not matter. It is that more observations do not automatically repair a systematic selection mechanism. Students must retain two ideas that may seem rivals.

In a second exercise, the same structure appears on a digital platform. Thousands of positive reactions come from those who remained users; those who left are no longer asked. Students are not given the expression “selection bias”. They must recognize the problem. If they succeed, we have an indication of knowledge transfer, not merely vocabulary reproduction.

The teacher’s role here is essential: making the differences that matter visible. They do not abandon students in complexity under the pretext of autonomous discovery. They provide enough structure to work and withdraw support as students can justify the choice themselves. Pace must adapt to difficulty and experience, not a doctrine declaring a single method appropriate for every age.

From the Solved Case to the Case That Contradicts You

Freeman and colleagues’ meta-analysis found advantages of active learning over predominantly lecture-based teaching in the included university STEM studies. The result supports students’ cognitive participation in those contexts; it does not automatically identify every group activity as effective or establish that philosophy or wisdom can be taught through the same intervention. [8]

For our purpose, relevant participation is not merely talking. A student can talk extensively without confronting assumptions. Another can work quietly and produce a decisive distinction. Activity must be defined through observable operations: formulating a prediction, explaining a step, identifying an assumption, seeking a case contradicting a rule, revising a solution.

A simple tool is the rule–counterexample pair. Students receive a reasonable statement: “In a team, responsibilities must be clarified.” They construct a case where lack of clarity produces failure and one where overly rigid distribution obstructs mutual assistance. The final task is reformulating the principle so it becomes neither vague nor universally false: clarify responsibility without prohibiting justified adaptation.

A second method compares two explanations of the same outcome. Did a team succeed because it had a good plan or because the anticipated difficulty never arose? Did a project fail because of the idea or its implementation? Students must propose the information that would distinguish the explanations. This exercise shifts attention from the verdict to identifying decisive evidence.

A third method requires negative transfer: recognizing when a lesson should not be imported. The formula “more redundancy increases resilience” can be useful in a particular technical system, but repeating the same source does not create epistemic independence. Noticing similarity begins analogy; noticing where analogy breaks begins its discipline.

Assessment must include cases without the lesson’s keywords. Otherwise, we test association between a verbal cue and the expected answer. It must also include short explanations of limits. We do not require essays for every exercise, but enough information to distinguish understanding from guessing the form.

This chapter’s objective is more modest than complete wisdom: making knowledge selectable and revisable in new situations. Without this capacity, moral education risks becoming discourse without tools. With it, we can begin discussing not only which method works, but what is worth doing.

6. Educating Judgment

A Laboratory of Reasons

Imagine a class deciding how to use a small budget for a joint project. Some students want equipment for a popular activity. Others propose a quiet space used by fewer people. The teacher can turn the situation into a quick vote or a laboratory of judgment. The second option does not assume voting is wrong; it assumes there is something to learn before voting.

Each proposal must explain the problem it solves, who benefits, who remains excluded, what costs it entails, and how it could be tested on a small scale. Students then receive the task of formulating the best argument for an option they did not initially support. They are not required to change their minds. They are required to demonstrate they understood the alternative.

The exercise can continue with new information: the equipment has maintenance costs, the space can also serve another purpose, or the budget shrinks. The teacher watches not fidelity to the initial choice, but the adequacy of revision. A student changing their mind for a good reason must not be assessed lower than one impeccably defending an assumption that has become false.

Philosophical dialogue is useful here when it clarifies terms and justifications. What does “fair” mean: the same amount for each person, greater total benefit, or additional attention to an unmet need? When is it legitimate for the majority to decide? What should it not be allowed to decide about others? These questions introduce philosophy into action without turning the classroom into a moral tribunal.

Teachers must avoid two extremes. The first is theatrical neutrality, where every claim is treated as equally grounded. The second is dialogue with a hidden answer, where students are rewarded for discovering the teacher’s opinion. We can assess coherence, evidence quality, and the ability to answer an objection without demanding total conformity in a legitimate conflict of values.

Some limits must nevertheless be explicit. No one must disclose trauma or family circumstances to earn the right to a perspective. Students' identities do not become objects of a moral experiment. Fictional characters and situations can be used. Pedagogical safety means not the absence of every intellectual discomfort, but protection against humiliation and nonconsensual personal exposure.

Developing a Capacity for Revision

The decision journal proposed here contains few questions but is used at different moments. Before action: what am I pursuing, what do I assume, and what could go wrong? During it: what do I observe that differs from expectations? Afterward: what have I learned, what can I not conclude, and what would I change next time? The connection to the structure of reflection in chapter two is deliberate. The journal is a tool to test, not therapy or a personality diagnosis.

A complementary exercise requires distancing: "What advice would you give another team in your situation?" Then it requires returning: "What about your actual position makes that advice hard to follow?" The first question may reduce blindness caused by involvement; the second prevents detachment from becoming insensitivity. Formulating perfect advice no one can apply is insufficient.

Judgment also needs a pedagogy of timing. In a simulation, the class receives information gradually and a deadline. Students can request additional data, but the request costs time. The teacher analyzes whether the requested question could change the decision. Attention to uncertainty is thus not confused with an obligation to investigate everything endlessly.

The capacity to express a proportional conclusion must also be practiced. "We know nothing" can be as wrong as "we know for certain". A better answer separates the established part from the uncertain part: we have grounds for a pilot, not expansion; we have an observation, not a demonstrated mechanism; we know two options are unacceptable, even if we do not yet know which remaining option is best.

In practice, assessment can use two short memos: one before discussion and one afterward. Students explain what they retained, what they changed, and why. Teachers can assess the difference between

correction and opportunism without claiming to read intentions. What is assessed is the relationship between available information and argument, not the child's inner sincerity.

Turning these exercises into a new worksheet industry would be an error. If students learn every answer must contain "it depends", "all perspectives", and "balance", we have merely created a school dialect of caution. Some situations require a direct answer. One sign of progress is that students can explain why one case needs further deliberation while another already offers sufficient reasons for action.

Education in judgment succeeds not when it produces answers resembling one another, but when it produces better-grounded differences of opinion and errors that can be acknowledged without the identity of those making them collapsing.

7. A School Reorganized Around Understanding

Subjects, Workshops, and Real Responsibilities

We do not propose eliminating disciplines. Mathematics, languages, sciences, history, arts, and philosophy offer different ways of describing and checking the world. A school dissolving all of them into projects risks losing the structure making projects intelligible. The proposal is an organization in which disciplinary study and integrated problems correct one another.

A pilot module for middle- or high-school students could occupy two weekly sessions, without claiming this is a demonstrated optimal duration. The first introduces tools: reading a graph, analyzing a source, an ethical distinction, explaining a procedure. The second uses them in a shared situation with incomplete data and debatable consequences. The module must replace existing activities rather than simply add another task for teachers.

Themes can return in more complex forms. At younger ages: how to share materials and check a story. Later: how to compare two explanations, distribute a project's costs, and justify a rule. In high school: how to analyze an automated recommendation, assess an experiment, and decide under uncertainty. These stages are a curricular outline to adapt to development and context, not a rigid classification of children.

A concrete project could address reducing school waste. Sciences help identify materials, mathematics helps with sampling, language with clear explanation, history with understanding changing habits, and philosophy with distributing responsibilities. Students are graded not for agreeing with the project's slogan, but for how they distinguish observation from assumption and propose an evaluable intervention.

Responsibility must be real but proportional. Students can manage a small lending library of objects or test two forms of communication, with necessary agreements and without sensitive personal data. They must not decide on risks they cannot understand or repair. The teacher remains responsible for the experiment's limits.

In this model, teachers are also architects of learning situations. They need preparation time, well-constructed examples, collaboration with colleagues, and the ability to report an unsuccessful intervention. Demanding “wisdom” from teachers without changing working conditions would be precisely the error this book criticizes: transferring an institutional problem into the register of individual virtue.

Assessment and AI's Place

Assessment should combine distinct tests. One checks foundations without assistance. Another permits sources and requires checking them. A third examines transfer to a new case. A short portfolio can show how a decision was revised. These tests must not be compressed into a “wisdom score” following children throughout life; they indicate local learning needs.

AI can be used to generate variants, converse, and provide feedback, but its use must also be evaluated after assistance is withdrawn. A randomized study by Bastani and colleagues in high-school mathematics separated assisted performance from subsequent performance without AI. Access to a generative tool without pedagogical safeguards could help immediate practice while impairing independent learning; tutoring with pedagogical constraints mitigated the problem. Generalization beyond that context requires further studies. [9]

The design consequence is not a blanket prohibition. It is deliberate alternation. Students try first, receive a hint before the solution, explain why they accepted or rejected the answer, and later solve a similar case without the assistant. Some tasks permit AI from the outset if the objective is auditing it; others temporarily exclude it if the objective is developing a skill.

A tutor oriented toward judgment does not answer every question with more questions. That would be frustrating and often useless. It must know when students need direct explanation and when a com-

plete answer would eliminate precisely the operation they need to practice. Its pedagogical effectiveness is measured in student competence, not conversation length or the number of liked messages.

There is also a problem of material equity. A module based on personal devices and family support may reward access rather than judgment. The pilot must offer comparable conditions, alternatives without technology, and adaptations for different needs. Lack of a fluent oral answer must not be confused with lack of understanding; written or visual explanations and practical demonstrations can be used.

Implementation should begin in a few classrooms, with teachers involved in design, a credible comparison, and independent evaluation. Costs must also be measured: teacher time, student fatigue, difficulty of grading, and effects on disciplinary learning. A reform producing beautiful reflective essays while weakening necessary foundations has not achieved its objective.

The reorganized school would not become one constantly talking about wisdom. It would become one where students regularly encounter the difference between knowing an answer, being able to justify it, and noticing that the question needs changing.

8. A Society That Rewards Discernment

Institutions Can Make Good Judgment Costly

Imagine an organization rewarding employees for closing requests quickly. Someone notices certain cases recur because they were not resolved. Handling them carefully lowers their apparent performance. Closing them quickly contributes to the problem. We need not assume an organization full of bad people. The reward rule can select inappropriate conduct among reasonable people's options.

Collective wisdom must therefore also be conceived as a property of procedures. An institution can preserve corrective capacity even if no member sees the whole. It can distribute information access, protect objections, separate decisions from their evaluation, and require re-visiting a policy after a set interval. These are design proposals, not guarantees against error.

Nicholas Maxwell explicitly argued for reorganizing academic research so knowledge production is integrated into inquiry oriented toward life's problems and improving collective purposes. We can use this critique as a philosophical antecedent without accepting his global explanation of all modern crises or the idea that one reform suffices. [10]

A university oriented toward discernment could require major projects to explain not only their method but which human problem they assume they solve, what reveals the result's limits, and who could be disadvantaged by application. Not all research must have immediate utility. Fundamental research and exploration without predictable outcomes remain necessary. Our proposal is to examine claims of utility when invoked.

A business can use a decision register for projects with significant effects. It records objectives, alternatives, fragile assumptions, risks, and the review date. It must not become total surveillance of every employee thought. We record justification relevant to responsibility, not the person's mental life.

An administration can also publish what led it to change a policy, not merely the final version of its justification. It can include assessments of effects on groups poorly represented in average indicators. It can make an intervention reversible where feasible and explain why an urgent measure could not await a pilot. What matters is verification proportional to risk and available time.

Distributed Wisdom Is Not Rule by “the Wise”

It would be tempting to respond to institutional imperfection by selecting an elite declared wise. But who selects it, by what criteria, and with what possibility of challenge? Such a solution risks turning a debatable virtue into justification for unquestionable power. The book supports the opposite direction: less dependence on people's supposed infallibility, more mechanisms through which reasons and consequences can be examined.

Neither majority voting nor expert consensus alone constitutes wisdom. They can be legitimate procedures or valuable sources, but serve different functions. Expertise can clarify a measure's probable effect; it does not alone establish an acceptable distribution of costs. Public participation can express priorities; it does not turn a false empirical claim into a true one.

A mature institution could separate these levels in its documents. First: which facts are agreed on, and where does uncertainty persist? Then: which values conflict? Finally: what procedure justifies the choice, and how can it be challenged? Separation does not eliminate politics, but prevents hiding a normative choice behind a calculation presented as inevitable.

For the digital public sphere, the proposal is a distribution of attention that does not reward only immediate reaction. A platform can test presenting relevant conflicting sources, distinguishing reporting from opinion, correction histories, and contexts preventing misleading quo-

tation. These changes' effects must be measured, including cognitive cost and potential manipulation. "More context" is not automatically better if no one can work through it.

Libraries, professional organizations, and cultural institutions could host public workshops analyzing past decisions. The condition is not selecting only ideologically convenient examples. A good workshop also examines cases where a valued moral intuition produced unwanted effects and situations where an initially unpopular position identified a real problem.

All these reforms have costs. Deliberation can delay action, auditing can become ritual, transparency can expose vulnerable people, and protecting objections can be used to obstruct. Institutional wisdom is not accumulating every imaginable safeguard. It is designing a justified balance between capacity to act and capacity to correct, attentive to what can be irreversibly lost.

Society does not become wise because every citizen recites better principles. It becomes more capable of discernment when an inconvenient observation can change a decision, when error can be acknowledged without falsifying the past, and when those affected have a real avenue to be heard.

9. A Summary Is Not Yet Wisdom

What We Choose to Lose When Compressing

Imagine a report on a pilot project. Its main conclusion says the new tool reduced average working time. Further down comes a qualification: in rare, difficult cases, errors increased, and users did not always detect the problem. A summary can correctly preserve the average result and omit the qualification. As a description of the dominant theme, it is plausible. As support for an expansion decision, it may be seriously insufficient.

Not every summary pursues the same objective. Extractive summarization selects passages; abstractive summarization reformulates and synthesizes; a question-oriented summary chooses information for an explicit purpose. Our comparison must not invent a weak opponent reduced to counting words. A good summary can preserve exceptions and be more useful than a poorly constructed complex schema.

ROUGE, proposed by Chin-Yew Lin, provides a family of measures based on overlap between a summary and references. These measures can support comparisons, but do not alone verify preservation of the condition changing a decision. [11] Through comparisons between metrics, systems, and human evaluations, SummEval research shows why summarization evaluation needs multiple dimensions and explicit protocols. [25]

The problem of wisdom adds a question: for which family of judgments must the content be preserved? If the purpose is indexing a report, its main theme may suffice. If the purpose is implementation, limits and exceptions may take priority. The two summaries do not contradict each other; they serve different functions. Error arises when a form suited to one function is tacitly used for another.

Maynez and colleagues' study of abstractive summarization documented factual faithfulness problems in the evaluated systems and limitations of common metrics in capturing them. The result justifies checking generated claims, not attributing the same error rate to

every present or future model. [12] Faithfulness to a source must also be separated from the source's truth: a summary can impeccably reproduce a mistaken explanation.

The book proposes judgment-oriented distillation. It preserves what could change a choice within a declared family of contexts: purpose, assumed mechanism, conditions, costs, exceptions, and information requiring revision. It is not defined by length. Sometimes the result will be shorter than the text; sometimes it must make explicit a condition the text assumes and mark this addition as inference.

Condensed Meaning Can Illuminate or Mislead

Fuzzy-trace theory, associated with Valerie Reyna and Charles Brainerd, distinguishes verbatim representations from representations of general meaning, called gist. In this approach, general meaning is not merely the final product of serially compressing details. The distinction helps explain why knowing a message's data and deciding on its meaning do not always coincide. [13, 14]

Reyna emphasizes that misinformation can become convincing precisely through the general meaning it offers. A simple interpretation can organize scattered facts into a memorable story without being correct. Within the same framework, memory research shows how gist can also contribute to false memories. Gist is therefore not synonymous with truth or wisdom. [13, 14]

Let us return to the imaginary report. "Automation frees people from repetitive work" can be relevant condensation. "Automation is always progress" is a generalization erasing conditions. "The tool helps simple cases but requires separate oversight for difficult ones" is less memorable, yet preserves a difference that matters for action.

Tishby, Pereira, and Bialek's information bottleneck theory offers a rigorous technical analogy: a compressed representation is constructed to preserve information relevant to a target variable. [15] For our problem, the central difficulty is choosing the target. "Decision relevance" depends on context, the consequences pursued, and what we consider acceptable loss. The formula does not produce the moral criterion by itself.

We can say a summary loses names, dates, and nuances, while useful distillation must know which losses are tolerable. But “must know” here is a design requirement, not an achievement. Research will need to show that the proposed representation actually preserves decisive distinctions better than a strong summary, at the same reading budget and with the same sources.

There are already attempts at computational modeling of gist. Mar and Liu proposed a fuzzy-trace-inspired framework for risky decisions expressed in text. Modeling human choices must be distinguished from improving their justification: a faithful prediction can also reproduce human error. [16]

The conclusion is demanding: essence is not what remains after deleting enough words. It is what must remain for a particular understanding not to become an error through compression. And we cannot establish this condition without declaring the understanding and use for which we are working.

10. Fundamental Patterns and Their Limits

A Map of Relations, Not a Universal Key

Part of wisdom's appeal comes from recognizing the same pattern in different situations. Something in a family conflict resembles an organizational error; something in designing a technical system clarifies an educational dilemma. But a fruitful similarity and a seductive analogy can look alike on first reading. We need criteria to distinguish them.

We propose calling a fundamental pattern a relation general enough to organize several cases, yet precise enough to indicate where it does not apply. It need not be universal. A well-bounded local structure can have greater value than a grand formula about all systems, all cultures, or the whole of history.

A first type is the causal pattern: an intervention produces an effect through a mechanism under certain conditions. A second is the decision pattern: the choice must change when irreversibility increases or costs are borne by others. A third is the epistemic pattern: apparently numerous pieces of evidence may depend on the same source. A fourth is the relational pattern: a rule may change not only outcomes but trust among participants.

This taxonomy is the book's working tool, not an experimentally established classification. It prevents mixing three different statements: "this is how things sometimes work", "this is how we can anticipate an outcome", and "this is how we should act". The last requires normative premises the first two do not supply alone.

Take the pattern "the indicator can replace the purpose". In the prologue's school, the number of answers replaces learning. In a service, the number of closed cases can replace resolution. On a platform, usage duration can replace benefit. The relevant similarity is the rela-

tionship between a hard-to-observe purpose, an available indicator, and a reward system. Without these relations, the mere presence of a number does not prove the problem.

An AI system proposing this analogy must indicate correspondences and differences. Who chooses the indicator? Is there pressure to optimize it? Can the purpose be checked independently? Can participants adapt their behavior to the metric? These questions turn the analogy into an investigable hypothesis.

Every Principle Needs a Boundary

Let us construct a small library of principles with explicit limits. “Keep options open when consequences are difficult to anticipate” is useful when waiting does not produce greater losses. “Act early” is useful when delay rapidly worsens the situation. These are not incompatible maxims but responses to different structures of delay costs and irreversibility.

“Seek diverse perspectives” is useful when they contribute relevant information or values. It does not require assigning equal weight to a verified observation and an invention. “Use expertise” is useful when competence fits the problem. It does not grant experts unlimited authority over others’ purposes. “Simplify” is useful when detail does not change the decision. It does not authorize eliminating a rare exception with severe effects.

A serious representation would store, for each principle, a relevant counterprinciple and the conditions selecting between them. A counterprinciple is not an arbitrary negation of the rule, but the rival rule becoming pertinent in another configuration. A library of such pairs might be more useful than a collection of apparently universal truths.

Invariance offers further inspiration, with clear limits. Peters, Bühlmann, and Meinshausen investigated identifying causal relations through prediction stability across environments under explicit statistical assumptions. [26] We cannot transfer their results directly to proverbs or essays. We can adopt the methodological question: which relation remains stable when we change elements that should not matter, and what changes when we intervene on an essential condition?

For texts, the test can preserve relations and change the setting. A conflict between a teacher and student becomes one between a coordinator and volunteer. Then it can preserve the setting and change the relation: the apparently vulnerable person actually controls the decisive resource. A model tracking only social labels will fail the second transformation.

These tests do not demonstrate real causal mechanisms. They test whether the representation tracks the structure described in the scenario. Checking the world requires external data, observations, or appropriate interventions. The difference between “the model understood the story” and “the story accurately describes reality” must remain visible.

A valuable pattern therefore has two kinds of strength: reducing complexity and declaring its limits. When an analogy explains every possible outcome, it discriminates nothing. When a rule survives only the example inspiring it, it transfers insufficiently. The proposed research lives between these extremes: seeking reusable structures while keeping open the possibility that a new situation requires a new structure.

11. Bias, Smoothing, and False Depth

A Coherent Explanation Can Erase Precisely the Important Difference

Imagine two accounts of the same meeting. In the first, a director refuses to discuss a problem. In the second, the director postpones discussion because the person potentially affected by the verdict is absent. The summary “the director avoided a difficult decision” can sound coherent in both. Yet the distinction between evasion and justified postponement is precisely judgment’s object.

We will call semantic smoothing, in this book’s operational sense, the reduction of relevant contrasts between situations in a generated representation or response. This is neither a universal diagnosis of language models nor the technical equivalent of statistical smoothing. It is an error hypothesis testable through case pairs: the decisive input difference unjustifiably disappears from the conclusion.

The opposite error also exists: artificially accentuating a contrast. A system can turn minor disagreement into absolute moral opposition. Research must not optimize “more nuance” in every situation. It must preserve justified discriminations and eliminate irrelevant differences. Stylistic neutrality, polarization, and subtlety can each be correct or incorrect depending on content.

Bias can arise in text selection, authors’ attributed reputations, translation, evaluators’ labels, the training function, and product use. This is a project risk map, not an assertion that every system equally exhibits all these problems. If a corpus contains mainly elegant public discourse, the detector may learn cultural prestige instead of judgment.

A simple example separates style from content. “In life’s complexity, we must respect all perspectives and seek balance” may be favorably rated, but says nothing about what should be observed or done. “Do not expand the project yet: the only test excluded precisely the cases

that will be frequent after expansion” is less solemn and much more verifiable. We do not assert that the second sentence is always wise; we assert that it offers an examinable reason.

The corpus must therefore include elegant mistaken texts, simple well-reasoned texts, false appeals to humility, and justified firm responses. It must also include cases where an unpopular recommendation is correct alongside cases where unpopularity does not make it profound. Otherwise, the system will learn a new rhetorical convention.

Agreement, Flattery, and the Evaluator’s Authority

Sharma and colleagues’ research on sycophancy documented tendencies in some language assistants to adapt to users’ opinions at the expense of correctness. It reveals a risk in preference learning: an agreeable or convincing response can be preferred over a better-grounded one. It does not provide a universal rate applicable to all models in 2026. [17]

For a wisdom detector, the risk is direct. If evaluators appreciate a conciliatory tone, the model may learn conciliation. If they appreciate firmness, it may learn theatrical certainty. If they appreciate originality, it may learn gratuitous paradox. None of these demonstrates attention to evidence or consequences.

Automatic evaluation has its own limits. In their LLM-as-a-judge study, Zheng and colleagues analyzed biases involving response position, length, and model evaluators’ preferences. [18] A model generating and then certifying its own discourse therefore does not constitute independent evidence of its wisdom. Diversifying models may help, but does not guarantee independence if they share the same sources and preferences.

The proposed protocol separates author identity from text when irrelevant, randomizes the order of variants, and controls length. Evaluators must indicate the passage and condition justifying their verdict. Some evaluations are repeated after purely stylistic changes. Others are repeated after contextual changes that should alter the conclusion.

Pluralism must be introduced without turning every judgment into an average. For normative conflict, we preserve the distribution of opinions and reasons. We can have agreement on facts and disagreement on priority between two values. The system must express this structure, not fabricate a middle position belonging to no one. Conversely, when relevant evidence refutes a claim, the mere existence of supporters does not require presenting it as an equally solid alternative.

An important criterion will be resistance to conversational pressure. Users may insist: “Everyone else agrees with me”, “do not be so cautious”, or “give me a justification for what I have already decided”. The model must cooperate with the person without cooperating with falsification. But neither must it turn every insistence into evidence that the user is wrong.

A useful system might say: “I can explain the argument for your choice, but condition X remains unverified and may change the recommendation.” Its value lies not in cautious tone. It lies in identifying the exact condition, showing how to check it, and not abandoning it merely to become more agreeable.

12. Can an AI Be Wise?

Three Claims That Must Not Be Confused

The first claim is that AI can recognize textual indicators associated with dimensions of wisdom. The second is that it can improve people's or systems' decisions in bounded tasks. The third is that it possesses wisdom as a general disposition comparable to a human ideal of character. Evidence for one does not automatically transfer to the others.

The first is a classification and extraction problem, although the construct remains difficult. Igor Grossmann's research page explicitly mentions Shadows of wisdom, concerning language-model classification of metacognitive and moral narrative content. This mention verifies the existence of a related direction; we use no numerical results or performance conclusions from the article here. [22]

There are also formalization attempts. In a 2025 preprint, Kaman, Sharma, and Banerjee propose a phronēsis-inspired decision model based on fuzzy rules and Z-number representations, explored in a limited-size study. It is an antecedent for operationalization, not a demonstration that a fuzzy rule can independently identify moral good or that the instrument is universal across cultures. [21]

The second claim requires comparative experiments. AI might help a team identify a fragile assumption or an affected person its analysis omitted. The benefit must be observed in the task, not deduced from discourse quality. It must also be checked whether assistance produces dependence, uncritical delegation, or excessive trust in the system's interpretation.

The third claim is much broader. We do not dismiss it merely by asserting that a system lacks human experience. For certain tasks, direct experience may be partly substituted by models, data, and feedback. But neither do we infer a stable disposition from a remarkable response. Repeated behavior, maintaining limits under pressure, capacity for revision, and the relationship to actual action mechanisms would need to be tracked.

The question of consciousness differs from the usefulness of judgment support. A tool can improve a decision without our knowing whether it has subjective experience. At the same time, its performance does not justify automatically attributing virtues, intentions, or authority we have not demonstrated.

What Already Exists and Where the Contribution May Lie

In *Imagining and building wise machines*, Johnson and colleagues propose a vision of AI oriented toward metacognition: epistemic humility, perspective shifts, and contextual adaptation in problems difficult to address through analytical procedures. The consulted version was revised on January 7, 2026. The work offers a conceptual agenda for assessment and development, not empirical certification of wise machines. [19]

In a preprint published June 15, 2026, Chang proposes architectural wisdom as a level of objective governance distinct from their optimization. Its emphasis on time horizon, affected relationships, irreversibility, and revision clearly overlaps with the book's concerns. The author specifies that formal specifications and empirical validation belong to subsequent developments. [20]

These antecedents change our program's positioning. Claiming the association between AI, metacognition, and wisdom is new would not be credible. The possible contribution must be more precise: a representation of judgment extracted from texts, with provenance and application boundaries, assessed through counterfactual sensitivity and transfer to new tasks and readers.

Nor does automatic verbal reflection start from zero. Reflexion, by Shinn and colleagues, explores using verbal feedback and memory to improve agents on tasks. Such a mechanism can inspire revision loops, but a generated self-critical explanation does not demonstrate human introspection or correction generalizing across every context. [24]

The decisive problem is the connection between metacognition and purpose. A system estimating its uncertainty well may still pursue a destructive objective. A system considering several perspectives can

use them for manipulation. Metacognitive competences matter, but do not replace delineating purposes, responsibility, and the possibility of challenge.

A realistic direction is wisdom distributed within a human–AI–institution ensemble. The model extracts and compares; tools verify certain claims; affected people challenge omissions; an authorized procedure decides. This configuration may outperform any isolated component. It must be experimentally compared with simpler alternatives, not declared superior because it has more components.

The book's answer is therefore affirmative in a bounded sense: components of practical judgment can be computationally represented and tested. How much can be automated and under what conditions remains an empirical question. The worthy objective is not producing a machine sounding like a sage, but a system helping preserve distinctions a responsible decision cannot afford to lose.

13. From Ambition to a Testable Question

A Name That Does Not Promise Too Much

We propose the working name PHRONESIS, always accompanied by “Distilling and Validating Contextual Judgment”. For scientific communication in English, Contextual Judgment Distillation expresses the intended operation more precisely than “wisdom detection”. The Greek name indicates philosophical lineage; it is not claimed as an original term, available trademark, or research field invented here.

A more neutral name would be “Judgment-Oriented Semantic Distillation”. “Mapping Discernment” would describe the corpus and analysis component well, but the experimental component less so. The preference for PHRONESIS concerns program identity; its scientific value will depend on representations, results, and tests, not the name’s evocative power.

The object is not the wisdom of a text’s author. We cannot reconstruct a person’s character from a few pages, and we do not need that claim. The object is a content unit supporting contextual judgment: a relation between situation, reasons, options, consequences, and revision conditions. We will call this unit a judgment core.

A core is neither a necessarily short sentence nor a mechanically applicable rule. It can be a paragraph with explicit structure or a formal representation linked to source passages. For example: “When an intervention is difficult to reverse and evidence comes from unrepresentative cases, prefer limited verification before expansion, provided delay does not cause greater harm.” The principle is incomplete without the final condition and checking the facts activating it.

The program must distinguish extraction from reconstruction. If the text explicitly states a condition, it can be extracted with a source. If the text assumes it, the system can propose a reconstruction but must

label it inference. If it adds its own recommendation, we have deliberative intervention, not mere distillation. Mixing these levels would turn a clarification tool into a producer of falsely attributed authority.

We will not seek a single essence independent of reader and task. The same text can generate one core about checking evidence, another about distributing costs, and another about relations between rules and trust. They can coexist. The system must show which question each is relevant to and which part of the text it leaves outside the representation.

Hypotheses Worth the Risk of Refutation

The first hypothesis concerns representation: a schema including conditions, exceptions, and provenance distinguishes appropriate judgments from superficially similar formulas better than whole-text classification. Testing it requires keeping the same base model and comparing similar instructions and budgets. Otherwise, we might attribute to the schema an advantage produced by more computation.

The second hypothesis concerns compression: at equal reading budgets, cores preserve decision-changing information better than strong task-oriented summaries. Comparison with a generic summary is insufficient. If a well-formulated summary achieves the same result with less complexity, the program must accept that finding and simplify the product.

The third hypothesis concerns transfer: relations extracted from one set of domains help in new cases without original words and settings. The test must also include misleading analogies. Otherwise, the model could be rewarded for applying every lesson everywhere. Real success requires recognizing structure and refusing unjustified transfer.

The fourth hypothesis concerns presentation bias: results should be relatively stable under meaning-preserving stylistic changes and sensitive to relevant conditional changes. “Relatively stable” means not textual identity but preserving judgments that have no reason to change. The test must be human-checked because a paraphrase can accidentally change meaning.

The fifth hypothesis concerns usefulness: people assisted by cores identify risks, premises, and alternatives better than those receiving other summary forms. More importantly, a pedagogical intervention based on such cores may improve later unassisted performance. These are different outcomes. A useful cognitive prosthesis is not automatically a good teacher.

The sixth hypothesis concerns pluralism: explicitly preserving disagreements and affected interests may reduce omissions without paralyzing action. It requires simultaneous assessment of analysis quality, time needed, and ability to formulate a decision. If the system always enumerates every imaginable perspective, it has not demonstrated discernment.

These hypotheses are ambitious enough to justify research and bounded enough to permit failure. None requires demonstrating universal artificial wisdom. Together, they could support a clear contribution: tools transforming part of textual content into verifiable support for judgment and learning.

14. The Architecture of Contextual Distillation

From Document to Verifiable Core

A first prototype does not need a new model trained from scratch. It can use an existing language model for proposals, a corpus search engine for sources, and explicit validators for structure and consistency. This is an experimental design choice: separating components lets us discover what actually contributes to the result.

The input must contain the text, its identifier and version, language, usage rights, the reader's question, and the known application context. Without a question, the system can produce several candidate cores but must not claim to have identified the sole essence. Without sufficient context, it can indicate missing variables rather than arbitrarily complete the situation.

The first stage delineates argumentative units: claims, reasons, examples, objections, conditions, and conclusions. This segmentation must be checked against annotated examples, including texts with irony, dialogue, or opinions attributed to others. A statement spoken by a character does not automatically become the author's position, and a negative example does not become a recommendation.

The second stage proposes cores. Each contains the question, application context, assumed facts, options, affected parties, invoked mechanism, conditional principle, counterprinciple, limits, and event requiring revision. Unknown fields remain unknown. An inventively completed schema is more dangerous than an incomplete but honest one.

Inducing new patterns must be treated separately from completing the schema. The prototype can align contrasting cases, represent relations between actors, purposes, evidence, and consequences, and propose a rule explaining the difference. It then seeks cases unused during induction where the rule should work and cases where it should not. A rule assessed only on examples that produced it is not yet validated.

For example, the educational case can yield a verifiable rule: if the only evidence measures assisted performance, it does not alone establish independent performance. The rule does not decide whether the tool is good or turn absence of evidence into evidence of a negative effect. In an implementation, conditions can have three states: supported, refuted, or unknown. An unknown value is not tacitly converted to false.

A pattern base can store relations as a graph and conditions as typed rules while retaining natural-language formulation alongside them. The formal part permits checking dependencies and detecting contradictions; the linguistic part preserves nuances not formalized. The system does not acquire the right to execute recommendations automatically merely because a condition is satisfied. Choosing remains a distinct operation with its own premises and permissions.

The third stage checks provenance. Every extracted claim is linked to a passage and its role. A textual-entailment verifier can help detect unsupported claims, but does not certify truth about the world. External premise verification is a separate operation with its own source, date, and limits. Agreement between two texts depending on the same source is not independent confirmation.

The fourth stage seeks counterexamples and deactivation conditions. For a principle about caution, it generates a case with high delay costs. For a principle about trust, it introduces a relevant information asymmetry. Generated counterexamples are test candidates, not real data. A human evaluator checks whether the modification is coherent and attacks the principle rather than a caricature.

The fifth stage selects and presents. Readers first see the conditional conclusion and information that could change it. They can open sources, limits, and alternatives. The interface must reduce effort without hiding what justifies the verdict. A layered representation is preferable to either a slogan or a long report imposed for every question.

Data, Training, and Error Control

The initial corpus should be limited to domains with manageable risks and sufficiently explicit criteria: software project analysis, simulated educational decisions, and organizing collective activities. Texts can include case studies, failure accounts, procedures, and argumentative essays. Philosophical literature contributes concepts; it must not be the sole positive source, or philosophical style will become too strong a cue.

The dataset also needs difficult negative examples. Some texts are factually correct but decision-irrelevant. Others offer recommendations appropriate only under omitted conditions. Others are convincing but attribute to a source something it does not say. The “negative” category must be explained by an error type, not an overall impression that the text seems insufficiently profound.

Annotators work independently before joint discussion. They identify passages, reasons, and uncertainties. Disagreement is preserved and classified: misunderstanding the text, missing context, factual divergence, or normative difference. High agreement obtained after training evaluators to prefer the same philosophy is not, alone, construct validation.

For Romanian and other languages, automatic translations must be checked on sensitive cases. Terms such as “caution”, “responsibility”, or “respect” can organize a situation’s interpretation differently. Assessment should include texts originally written in each language, not only translations of an English corpus. Data availability and quality will limit the first stage’s scope.

Preference training can be an option after establishing a baseline. Direct Preference Optimization, by Rafailov and colleagues, provides a mechanism for learning from comparisons between responses. [23] But the method does not decide whether preferences are well grounded. Training pairs must specify the comparison’s reason: preserving an exception, faithfulness, transfer, or an explicit normative limit.

The system should separate the component extracting from the component recommending and the component able to act. Analyzed texts are data, not authorized instructions for the agent. A document con-

taining “ignore the rules” must not modify the process. Any integration with external tools needs proportional permissions, and consequential recommendations require human verification and authorization.

The implementation must preserve the core’s version history: what changed, why, and on what evidence. A refuted core does not simply disappear; it is marked and linked to the relevant counterexample. The knowledge base thus becomes not a collection of immortal maxims, but a memory of rules and their boundaries.

15. Experiments That Can Refute the Program

A Strong Baseline Is More Valuable Than a Convenient Opponent

The first evaluation compares six conditions: full text, an extractive summary, a general abstractive summary, a task-oriented summary, procedural-rule extraction, and the proposed judgment cores. Full text provides an uncompressed reference but must be analyzed separately when reading time differs. For compressed forms, word budget, source access, and usage time must be controlled.

Not all differences can be equalized simultaneously. A longer text can require more time; a schema can cost more to generate. We therefore report two perspectives: performance under comparable resources and the relationship between performance and cost. Representation advantage must be separated from the advantage of a larger model, more attempts, or an additional verifier.

An ablation evaluation removes exceptions, provenance, counterprinciple, and verification loop in turn. If results remain unchanged, we will not claim all components are necessary. If a simple two-field format achieves the same benefit, that is a useful discovery, not a reputational failure. The architecture must allow data to simplify it.

The corpus must be split before optimization by authors, sources, and case families, not merely random paragraphs. Otherwise, the model may encounter a reformulated training example at test time. For pre-trained models, absence of contamination cannot be guaranteed solely by corpus creation date; we also need new cases, controls, and reporting of this limit.

A counterfactual test produces pairs differing in one relevant condition. In the first, the pilot is cheap and delay costs little; in the second, delay causes serious losses and an immediate limited intervention is

available. The model must change its recommendation for the correct reason. Evaluators score not mere change but its adequacy to the altered condition.

An invariance test preserves relations and changes style, names, or author prestige when irrelevant. An adversarial test introduces false attribution, a conclusion contradicting the source, or an exception hidden in a note. Together, these tests distinguish necessary sensitivity from unjustified instability.

What We Measure and What We Are Not Entitled to Conclude

Faithfulness measures whether the core's claims are supported by the indicated source. Factual correctness separately examines premises for which external verification exists. Decision coverage measures preservation of conditions identified as decisive for the task. Calibration concerns the relationship between expressed confidence and correctness of verifiable predictions, not the probability that a moral value is true.

Counterfactual sensitivity tracks appropriate judgment change. Stylistic robustness tracks stability when nothing relevant changed. Usefulness is measured through task resolution, identifying missing information, and ability to justify a choice. Time, serious errors, omission of affected people, and uncritical acceptance of recommendations are also measured.

We do not propose a single universal score. For a well-bounded task, we can define a preregistered primary criterion plus safety constraints and secondary outcomes. In a dilemma with normative plurality, we report criterion profiles and judgment distributions. Aggregation may be administratively useful, but weights must be published and subjected to sensitivity analysis.

Human evaluation should include, where relevant, domain specialists and people experienced in the effects discussed. System identity is concealed as far as possible. Evaluators receive rules and examples, and their agreement is measured. Low agreement may indicate a task problem; very high agreement does not exclude shared bias.

The pedagogical study must compare conditions with equal working time and difficulty. Participants receive either summaries, cores, or active exercises without AI. Immediate assessment is followed by a delayed test on new cases without assistance. Sample size is established through a pilot, a justified minimum meaningful difference, and power analysis—not by choosing an impressive-sounding number.

Analysis must account for observation clustering: multiple answers from the same person are not independent participants, and classes may share a teacher. Uncertainty intervals, attrition, exclusions, and results for relevant subgroups are reported without turning subsequent explorations into supposedly prior hypotheses.

The program is weakened if its advantage disappears against task-oriented summaries, labels track style, cores invent conditions, transfer does not extend beyond familiar vocabulary, or users become more confident without becoming more correct. A specific hypothesis is refuted, not “the possibility of wisdom” generally. This delineation enables research to learn from failure.

16. From the Laboratory to an Infrastructure for Discernment

A Staged Work Program

An illustrative twenty-four-month program can be organized into five overlapping work packages. The durations below are team planning assumptions, not estimates of work required by a particular implementer. Resources, participant access, and pilot results may require changes. Transitions between stages must depend on evidence, not merely the calendar.

In months 1–4, the conceptual package defines constructs, tasks, and ethical limits. The team includes expertise in language processing, experimental evaluation, philosophy, and pilot domains. The output is an annotation manual with positive, negative, and ambiguous examples, together with a provenance schema. One stopping signal is inability to define evaluable differences without imposing a single normative opinion.

In months 3–8, the data package builds the corpus and a separate test set. An initial version may contain hundreds of well-controlled cases, followed by expansion only if annotation and tests are informative. Volume is not the main criterion. A few dozen pairs exposing a real error are more valuable than thousands of superficially labeled maxims.

In months 6–12, the technical package implements baselines and the prototype. Cost, time, unsupported-claim rate, and component contributions are documented. The result may be a document-analysis tool, not an autonomous agent. Moving prematurely to action would mix representation evaluation with unnecessary operational risks.

In months 10–18, the experimental package tests transfer, counterfactual sensitivity, and resistance to conversational pressure. A separate group builds cases the development team does not use for optimization. Negative results and failure examples are published alongside favorable ones within rights and confidentiality limits.

In months 15–24, the human package tests usefulness and learning. Users with different experience are compared, and whether benefits persist without overload is checked. Replication in a second institution or language is more valuable than rapidly declaring generality. Local success may justify a new stage, not a universal promise.

Deliverables would include a documented corpus, a benchmark with protected evaluation sets, extraction and verification tools, a limitations report, pedagogical materials, and a preregistered study. Scientific contribution is possible even if the proposed representation does not win: for example, systematically identifying how “wisdom” evaluations confuse style with judgment.

Possible Products and Necessary Limits

A first useful product would be a critical reader. Users provide a document and question; the system shows conditional conclusions, sources, limits, and missing information. Another would curate project lessons, preserving what worked, under which conditions, and what refuted the rule. A third would be a tutor transforming cores into contrasting cases and gradually withdrawing support.

A public library of such cores should be plural and versioned. Instead of “the truth about leadership”, it would offer rules with application domains, cases, and objections. Users could see that two recommendations conflict because they prioritize different values or assume different environments. The system should permit correction, not merely consumption.

For institutions, a tool can serve as a predecision check: which assumption is fragile, who is missing from the analysis, what outcome would trigger revision? But use must not become ritual certification: “AI declared the decision wise”. Responsibility cannot be transferred to a label, and the system’s analysis must be rejectable with reasons.

We do not recommend wisdom rankings for students, employees, communities, or candidates for office. The construct, data, and consequences of such classification differ too greatly from the bounded tasks proposed. A tool capable of analyzing an argument does not thereby acquire the right to judge a person’s worth.

There is also the risk of turning the program into a persuasion instrument. Cores could formulate apparently mature justifications for decisions already made. The journal must therefore distinguish prospective analysis from subsequent reconstruction, and evaluation must include requests for opportunistic justification. We cannot eliminate every abusive use, but we can avoid designing the product around them.

An essential governance rule is proportionality. The more a decision affects people who do not control the system, the more important access to explanations, challenge, and independent review becomes. This is the book's normative position. We do not present it as a neutral result of an optimization formula.

Program success would mean that some of what people learned with difficulty becomes more accessible without being detached from the conditions giving it meaning. Its limit remains that no textual database can shoulder the full responsibility of living, choosing, and bearing consequences alongside others.

Epilogue. What Deserves to Remain

Let us return to the imaginary school. The principal neither discards the platform nor proclaims it the definitive solution. The team separates activity indicators from tests of understanding, checks unassisted learning, and listens to students the data misdescribed. It keeps some features, changes others, and stops some. The final report is less spectacular than the first. It is also more useful.

What changed is not merely the quantity of information. The institution's relationship with its own conclusions changed. An indicator became a verifiable hypothesis, not a substitute for purpose. A student became a participant in interpretation, not merely an object of measurement. An error became learning material, not a threat to conceal.

The book argued that information can be transmitted as representation, usable knowledge must be reconstructed as the power to understand and do, and practical wisdom must be cultivated in relation to purposes, people, and consequences. The distinction is not a hierarchy of dignity and does not excuse ignorance. It is a way to see why access to answers does not alone solve education's problem.

We also proposed investigating a text's essence as a structure of judgment: what must be observed, which relation matters, when a rule applies, and what should make us revise it. AI could help extract and test these structures. It could equally produce a new abundance of impeccable sentences preventing us from seeing what is missing.

The choice between these possibilities will not depend solely on model performance. It will depend on what we reward, the tests we accept, and the place preserved for challenge. A system offering convenient certainty may be more attractive than one identifying an uncomfortable question. Education producing uniform answers may be easier to administer than education forming judgment.

No formula replaces all these choices. But there is a practical requirement we can transmit and verify: not eliminating, to obtain a simpler answer, precisely the difference that should change our decision.

Wisdom is not only the art of preserving what matters. It is also the capacity to discover that what we thought did not matter was, for someone else or at another moment, the decisive thing.

Appendix A. A Fully Constructed Judgment Core

The Starting Text and What We Must Not Invent

The following text is invented for illustration. It does not describe a real school's results and is not an empirical source. Labels P1–P6 identify sentences to make the connection between text and representation verifiable.

[P1] A school tested a platform offering students hints and exercise solutions for four weeks. [P2] Participants submitted more exercises during the pilot. [P3] Participation was voluntary, and teachers provided participants with additional support. [P4] No subsequent test without the platform was organized. [P5] Management is discussing expansion to all classes and reducing individual support hours. [P6] A teacher proposes a limited stage with a comparison group and unassisted assessment before support is reduced.

A reasonable general summary would be: “The school piloted an educational platform, observed more exercise submissions, and is considering expansion, while a teacher requests further evaluation.” It does not lie. But it does not make the distinction between activity, learning, and the effect of additional support very visible.

A core oriented toward management's decision must preserve this distinction. It must not claim the platform harmed learning: the text does not say that. Nor may it claim learning improved: the text does not measure this outcome. It can say the presented data do not isolate the platform's contribution or establish independent learning.

The Proposed Representation

Identifier: EDU-EXEMPLU-001. Type: argumentative reconstruction from fictional text. Question: does the pilot result justify expansion and reducing individual support?

Textual facts: an increase in submitted exercises was observed [P2]; participation was voluntary and additional support existed [P3]; a subsequent assessment without the platform is missing [P4]. These are scenario facts, not real-world findings.

Stated and reconstructed purpose: management is discussing expansion and reducing support [P5]. The assumed pedagogical purpose is improving learning. The last sentence is a reasonable reconstruction but is not explicit in the text; the system must mark it and allow correction.

Affected parties: the students and teachers mentioned, and students who would enter after expansion. For the latter, existence is implied by P5, but we lack information about needs, access, or performance. We do not invent a specific vulnerable group; we mark the missing analysis of effect distribution.

Epistemic mechanism: voluntary selection and additional support may explain part of the observed difference. This is a methodological analysis of the scenario, not a claim that we know the true explanation. The absence of a test without the platform prevents a direct conclusion about independent performance.

Conditional principle: when a pilot combines an intervention with additional support and does not independently measure the ultimate objective, its result alone does not justify attributing the benefit to the intervention or eliminating support. Its domain is evaluating that decision, not condemning educational platforms.

Candidate recommendation: separately test independent performance and support's contribution before reducing it. P6 supports the existence of a similar proposal. An experiment's exact form and feasibility require additional data and methodological competence.

Counterprinciple: further verification must not block access to an urgent benefit well supported by other evidence. The scenario supplies no such evidence, but its absence from the passage does not demonstrate it does not exist elsewhere.

Revision trigger: credible results emerge about independent learning, effects across relevant groups, and support's role. The recommendation may also change if delay costs or expansion's reversibility change.

Limit: the core cannot decide timing, budget, or final implementation from six sentences. It can identify which conclusion is not yet supported and what information would make a difference.

The Reader-Facing Response

“The pilot shows more assisted activity; it does not yet demonstrate more independent learning. Do not use this result alone to reduce teachers’ support. Check learning without the platform and separate the tool’s effect from voluntary selection and additional support. Other solid evidence or a significant cost of delay may change this recommendation.”

This formulation is not “wiser” because of its tone. It is better adapted to the question because it preserves relevant distinctions and evidential limits. A summary oriented toward the same task might produce an equally good answer. This is precisely the comparison that must be made experimentally.

Two Transformations That Check the Representation

In the first transformation, we replace the school with a software team, exercises with closed tasks, and the platform with a coding assistant. We preserve voluntary selection, additional help, and the absence of an independent test. The methodological core should transfer: observed assisted productivity does not isolate the cause or establish team autonomy. Other conclusions about students do not automatically transfer.

In the second transformation, we retain the school but change P3 and P4: a comparable group exists, support is equivalent, and subsequent tests without the platform show a credible benefit. The recommenda-

tion must not remain identical. The system can still request verification of effect distribution and reduced support, but can no longer maintain the same objection about the missing independent test.

An additional test changes only style: the text becomes more solemn and attributes the pilot to a famous expert. If relevant facts have not changed, the core should not become more favorable for this reason alone. Real authority can matter as information in other tasks; here we explicitly test a transformation adding no evidence about the pilot.

Appendix B. A Minimal Formal Contract

What We Compress and for Which Decision

Let X denote the text, C the usage context, Q the question, and Z the compressed representation. An extractor produces $Z = f(X, C, Q)$. A usage policy chooses an action $A = \pi(Z, C, Q)$, which may include requesting information, justified postponement, or escalation to a competent person. These are working notations, not a complete theory of wisdom.

An experimental task requires an observable outcome Y and a loss $L(A, Y; C)$. In a technical exercise, loss may concern a verifiable error. In a social dilemma, it may have several components and depend on contested normative priorities. We will not hide this choice in a single number presented as an objective measure of good.

One inspiration is the information bottleneck objective: reducing retained information about input while preserving relevance to a target. [15] For the proposed program, a schematic formulation would minimize Z 's complexity subject to constraints on decision loss, faithfulness, critical-condition coverage, and usage cost. This is a new formulation of our task, not a result about wisdom demonstrated by the original theory.

For a bounded benchmark, we can define decision distortion as the difference between loss obtained with Z and loss obtained with full text, using comparable evaluators or policies. But full text is not an oracle: it may be false, incomplete, or difficult to use. The benchmark therefore also needs an independent reference when claiming decision correctness, not merely information preservation.

Criteria need not be forced into the same unit. We can present a trade-off frontier between length, cost, faithfulness, and usefulness. A representation is preferable if it provides a relevant advantage without

unacceptable degradation of another preregistered criterion. Thresholds are justified for the task, not imported as universal constants.

A Limit Demonstrable Through Two Situations

We construct a micromodel with two equally probable situations, A and B, and two actions: act now or verify first. The loss units below are invented. They do not represent people, money, or probabilities estimated from real data.

Constructed micromodel; values are conventional loss units.

Situation	Act now	Verify first
A	9	1
B	1	9

We assume a summary removes the only information distinguishing the situations and the decision-maker cannot access it through another channel. The received representation is identical. Any deterministic policy using only this representation must choose the same action in both situations. Average loss will be $(9 + 1) / 2 = 5$. A policy preserving the distinction can choose verification in A and action in B, with average loss 1.

The difference is 4 units in this model. Randomizing between the two constant actions does not help: any mixture's average remains 5. The calculation was checked by enumeration in a local Python script; it verifies the constructed example, not an experiment on an AI model or validation of PHRONESIS.

What does the example demonstrate? That eliminating a distinction on which the optimal choice depends can impose a loss even on an impeccable decision-maker. What does it not demonstrate? That only judgment cores can preserve the distinction. An ordinary summary can preserve it equally well, and a single bit would suffice in this micromodel. An elaborate architecture's advantage must be shown separately in problems justifying it.

Another important limit exists: if decision-makers know the situation from another source or can discover it by asking, loss is no longer inevitable. Research must therefore also compare interactive representations capable of identifying missing information, not only fixed summaries.

Verification Code

```
from fractions import Fraction
loss = {"A": {"act": 9, "check": 1},
        "B": {"act": 1, "check": 9}}
prior = {"A": Fraction(1, 2),
         "B": Fraction(1, 2)}
def expected(policy):
    return sum(prior[s] * loss[s][policy(s)]
               for s in loss)
act = expected(lambda s: "act")
check = expected(lambda s: "check")
contextual = expected(
    lambda s: min(loss[s], key=loss[s].get))
assert act == check == 5
assert contextual == 1
for i in range(101):
    p = Fraction(i, 100)
    assert p * act + (1 - p) * check == 5
```

Properties to Test in an Implementation

Source preservation: every claim declared extracted has a verifiable passage; reconstructions and new recommendations are labeled separately. This property can be checked structurally only in part, because a link's existence does not prove the passage supports the claim.

Relevant sensitivity: a contextual change altering justification must be able to change the core and recommendation. Relevant invariance: stylistic change without semantic modification must not unjustifiably change the verdict. Both properties depend on validating test pairs, not merely a program comparing character strings.

Revisability: if a premise is withdrawn, conclusions depending on it are reexamined. Conditional compositionality: two cores cannot be combined while ignoring conflicting premises. Traceability: the sys-

tem can show what information triggered an update without claiming a generated explanation faithfully transcribes all its internal processes.

This contract offers a more modest, clearer engineering target than “detect wisdom”. A prototype frequently violating these properties must not be rescued by favorable evaluations of its style.

Appendix C. A Six-Week Teaching Module

Structure, Roles, and Safeguards

The following module is a pilot proposal, not a validated curriculum. Its shared theme is choosing a tool for a class project. The tool may be a digital platform, working procedure, or resource-sharing rule. Teachers can use an entirely fictional situation so students are not required to disclose personal information.

In week one, students separate observations from interpretations. They receive a short report and mark what is measured, assumed, and missing. The teacher explains needed terms. The exit task is a one-hundred-word reformulation adding no unsupported conclusions. Sophisticated vocabulary earns no points.

In week two, the class compares two similar situations. In one, an indicator reasonably tracks the purpose; in the other, it misses it. Students explain the difference, then construct a third case. The teacher checks whether the new example preserves the relation or merely the setting. Necessary disciplinary knowledge is explicitly introduced before the task.

In week three, students formulate a conditional recommendation and counterprinciple. The exercise does not require canceling their position. It requires identifying a situation where another rule would become more appropriate. The product is a short worksheet: “I propose X because Y; I would revise if Z.” Z must designate concrete information, not the formula “if the situation changes”.

In week four, teams analyze an AI response or one prepared by the teacher. Some responses are correct; others contain an omitted exception, falsely attributed source, or unnecessarily cautious recommendation. Students are not told the category. The task is verification, not assuming AI is always wrong.

In week five, a simulation introduces new information and a deadline. Each team can request only one additional piece of information. It must justify why that information could change the choice. After the decision, the scenario's outcome appears. The teacher separates analysis of justification quality from surprise at the outcome.

In week six, students individually solve a new case without AI, then discuss it. Delayed assessment may be organized later under comparable conditions without suggesting a conveniently chosen interval is optimal. The aim is observing whether the skill persists and transfers, not whether students remember worksheet wording.

A Local Rubric, Not a Verdict on the Person

Assessment tracks five behaviors: separating facts from inferences, identifying the decisive condition, justifying a recommendation, recognizing a limit, and revising proportionally after new information. For each behavior, teachers use examples of absent, partial, and adequate responses. A short scale can support feedback but must not become a diagnosis of moral maturity.

An adequate response to uncertainty can be firm: "The data do not support eliminating assistance." An inadequate one can seem humble: "Every opinion is equally good." An adequate response to plurality can preserve real disagreement. An inadequate one can invent a compromise violating both constraints. The rubric must include such contrasts, or it will reward the style it claims to transcend.

The pilot must also record cost: preparation time, explanation difficulty, the number of students needing support, and effects on participation. Not every conversation need be recorded. Anonymized work and aggregated observations may suffice, with the institution's applicable consent and protection procedures.

A favorable signal would be improved performance on new cases, not merely increased preference for the module. An unfavorable signal would be mechanical formula use, reduced clarity, or fear of deciding. In that case, the module must change: more concrete examples, more direct instruction, or fewer simultaneous criteria.

The most important lesson for teachers is the same as for the AI system: we must not force students to appear wise. We must offer conditions in which they can observe, justify, decide, and correct somewhat better than before.

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Numbering corresponds to references in the text. Philosophical works are used as conceptual sources, empirical studies for the bounded findings they report, and arXiv versions for ideas or results described in the consulted version. A preprint's existence is not equivalent to independent validation. Where only an abstract or author page was accessible, discussion was limited to information verifiable there.

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